W6OTX | W6ARA | PAARA NEWSLETTER | K6OTA | K6YQT

VOLUME 77, NUMBER 2, February 2026

PAARA*graphs*

The Official Newsletter of the

Palo Alto Amateur Radio Association, Inc.

The Friendliest Club Around

Celebrating 89 years as an *active* amateur radio club—*Since 1937*

http://www.paara.org/

The Stanford AREDN Project: From Concept to Completion

Rachel Kinoshita (KK6DAC) and Mason Matich (KC3WNY)

AREDN® strives to provide the Amateur Radio Community with a quality solution for supporting the needs of high-speed data in the Amateur Radio and Emergency Communications field. The AREDN Development Team was formed February of 2015 by former members of the BBHNDev team interested in making mesh software work for the needs of Amateur Radio Operators and emergency networks. Rachel and Mason will discuss their experiences with the project from concept to completion.

This meeting will be a
Hybrid Meeting
Zoom and In Person

Time: Feb 6, 2026 07:00 PM Pacific Time

Please check https://www.paara.org for Zoom Details

Upcoming Events

Feb 6 Mar 6 Apr 3	PAARA General Meeting, 7:00 PM to ~9:30PM Zoom and In Person Meeting.
Feb 18 Mar 18 Apr 15	Board Meeting, 7:00 PM. Everyone welcome! Zoom Meeting, eMail President for details.

President's Corner

Well, it's that time of the year again. I'm sure that phrase means different things to different people. To me it has meant a particularly busy time at work that has kept me from being on the air much in January. Hopefully that's not the case for most of you. But it's also a good time to look forward to longer days and the activities that PAARA has planned for 2026. First up is Home Brew night at the March 6th meeting. With one month to go, hopefully you've been working on your projects and your show-and-tell "presentations". This is always a spe-

(President — Continued on page 3)

Question *of the* MONTH

What is the best first radio for a new ham?

A Simple 30-Meter Antenna Surprise at Pacificon

Mikko Sannala AB6RF

At the 2025 Pacificon, we added a 30-meter CW station for the first time, which meant we needed an antenna just for that band. The other bands were already covered by the tri-band beam on the tower and a wire vertical on the hotel roof for 40 meters, but 30 meters required something new.

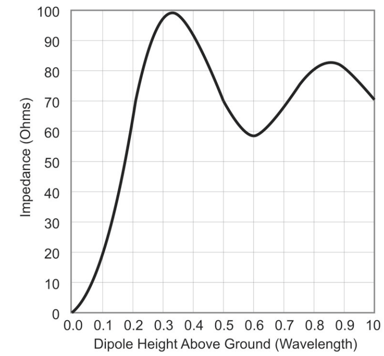
Since I didn't want to buy anything expensive, a wire antenna was the obvious choice. I tried a few ideas in an antenna simulator, including verticals and a delta loop, and eventually settled on a simple half-wave dipole. Hung roughly north–south, it looked like the best overall fit for our operating needs.

Anyone who has helped with antenna setup on Thursday knows that time is always tight. I wanted to avoid the usual cut-and-try routine, so I used the MMANA-GAL simulator to calculate the dipole length as accurately as possible ahead of time. While doing that, I noticed something interesting. Even if the antenna were cut perfectly, there was a good chance the SWR would still be higher than ideal simply because of how high the dipole would be above the ground.

This comes down to how antenna impedance works. The feedpoint impedance has two independent parts. One part tells us whether the antenna is too long or too short. If it's too short, it looks capacitive; if it's too long, it looks inductive. When the antenna is resonant, that part disappears. By trimming the antenna length, we can always make this reactive part go to zero, and that's the part most of us are used to adjusting.

The other part of the impedance is the resistive part, and for a simple horizontal dipole this depends mainly on the antenna's height above ground. Trimming the wire doesn't change it. In other words, you can cut the antenna perfectly, hit resonance exactly, and still end up with a poor match.

As a dipole gets higher above ground, the way it interacts with its reflection in the ground changes. At certain heights, especially around 0.3 to 0.4 wavelengths, the feedpoint resistance can rise well above 50 ohms. Based on our setup, we expected the 30-meter dipole to end up about 30 to 40 feet high, mounted below the tri-band beam. That put us right in that range. The simulator predicted a feedpoint resistance of around 90 ohms, which would correspond to an SWR close to 2:1.



Impedance vs Height

Of course, simulations are only simulations. The exact height, nearby metal from the tower trailer, and unknown soil conditions all affect the real result. We wouldn't know the true impedance until the antenna was actually in the air.

There was another wrinkle to consider as well. If you connect an antenna analyzer at the shack end of a coax cable, you are not measuring the antenna's feedpoint impedance directly. The coax itself transforms the impedance, depending on its length and frequency. To remove that effect, the analyzer has to be calibrated using that specific piece of coax, a process known as open-short-load (OSL) calibration. Once that's done, the analyzer shows the impedance at the antenna, not at the bottom of the coax.

To handle this, I brought along about 40 feet of coax that I had already calibrated with my antenna analyzer. On the setup day, we hoisted the dipole into position with that calibrated coax attached and took a measurement. The analyzer showed almost zero reactance, which confirmed that the antenna length was right on target. The resistive part, however, was 92 ohms — very close to what the simulator had predicted. I was actually pretty excited, not because the match was so bad, but because the prediction was so accurate.

At that point the problem was clear. The antenna was resonant and as high as we could reasonably make it, but the feedpoint resistance was too high. Trimming the wire wouldn't help, and lowering the antenna would only make things worse overall. Hoisting an antenna tuner to the top of the mast was not an attractive option either.

What we needed was a way to transform the impedance. One elegant solution is a quarter-wave

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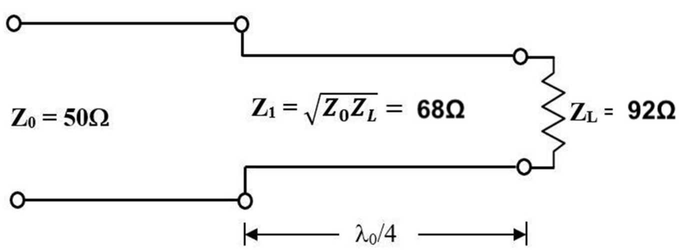
(Antenna Surprise — Continued from page 2)

transmission-line transformer. The basic idea is that an electrical quarter-wavelength of coax, with the right characteristic impedance, placed between the antenna and the feedline can convert one impedance into another. The math behind this is simple and, for once, actually useful. The characteristic impedance of the transformer section should be the square root of the product of the two impedances you're trying to match.

In our case, we were trying to match roughly 92 ohms at the antenna to 50 ohms at the rig. The square root of 92 times 50 works out to about 68 ohms. There's no such thing as 68-ohm coax, but 75-ohm coax is close enough to work quite well.



So we took the dipole down, removed the calibrated test coax, and connected about 16 feet of solid dielectric 75-ohm coax at the feedpoint, which is electrically a quarter wavelength on 30 meters. From there, we ran a regular 50-ohm coax back to the radio tent.

The result was an SWR of about 1.2:1 — just about dead-nuts perfect.

The big takeaway is that a dipole antenna can be cut perfectly and still be mismatched, simply because of its height above the ground. But with a little planning, and a short piece of the right coax, it's possible to solve the problem cleanly and efficiently. And every now and then, it's pretty satisfying when the real world agrees with the simulator.

PAARA 2025 Overview

On request, we will provide a copy of the club overview presented at the January PAARA meeting. You may email your request to secretary@paara.org.

(President — Continued from page 1)

cial night where club members get a chance to share information, experiences, and anecdotes about their various projects. Projects can be radio related, radio adjacent, or just fun. Each project gets about 5-7 minutes to present, so you don't have to put a lot of material together or be a polished public speaker. No matter what you've been working on, we'd love to hear about it!

Next up will be a PAARA in the Park event on April 18th. We'll be back at Memorial Park in Cupertino and look forward to a fun activity, BBQ, and QSOs, so save the date! It's hard to believe, but while Field Day is several months away in June, planning will be starting soon, and we expect to have some more extensive antenna work parties this year than we've had in the last few years. Keep an eye out for more information about those, and about how you can get involved in Field Day preparations. Everyone is welcome to chip-in however they can. In my experience, the preparation is a big part of the fun, so don't miss out!

I hope everyone enjoyed the year-end 2025 club review that was presented at the January meeting. Many thanks to the board for organizing and helping to prepare the review, and especially to our Secretary, Ric (N6AJS) and our Treasurer, Margaret (K6WEK) for their detailed work preparing the membership and financial information. For anyone that is interested, a copy of the presentation will be provided to any member who makes a request by emailing secretary@paara.org.

I recently received a dispatch from some of our PAARA members at Quartzsite (see photo on next page). One of these years I'll hopefully be able to join the fun myself, but in the meantime, we'll welcome them back at the February meeting and look forward to interesting stories, and hopefully an upcoming PAARAgraphs article about their adventures!

Lastly, I'd like to congratulate Doug Teter (KG6LWE), our newest recipient of a PAARA Lifetime Membership! If you attended the January meeting, you know that we were able to surprise Doug with this well-deserved recognition. If you didn't get a chance to congratulate him in January, give him a pat on the back the next time you see him. Doug has been and continues to be one of the people who keeps PAARA fun, active, and the friendliest ham radio club around! Congratulations, Doug!

I hope to see you all at the club meeting on February 6th, either IRL or on Zoom! If you're there in person, please join us after the meeting for pizza.

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(Continued from page 3)

Quartzsite Group: Doug, Rob, Michelle, Joel, Cathy, Jim, and Stiv

Just a reminder that we've moved to the Round Table Pizza at 702 Colorado Ave in Palo Alto – just a short drive north on Middlefield Road from Cubberly Community Center where the meeting is located. If you're not sure where it is or need a ride to get there, please ask at the meeting. We'd love for you to join us!

It's going to be a fun and exciting year at PAARA!

73, Bob KN6YGN
Get on the air to keep the airwaves alive!

Barista Opportunity

The club needs your help with setting up the coffee urns before the meeting starts, and with taking them down when the meeting is over. Simple and clear instructions will be available, as well as coaching as needed. If you can also store the two boxes between meetings and wash the baskets from the urns between meetings (particularly before a meeting where you can help with setup) that would be of great help, although it's not strictly necessary. Please add your name to the coffee rotation sign-up sheet which will be posted above the coffee urns, or get in touch with Sorin KN6YUH. Your help is greatly appreciated!

Jan 2026 Board Meeting Minutes

Present were President Bob Ridenour KN6YGN, Vice President Rob Fenn KC6TYD, Secretary Ric Hulett N6AJS, Treasurer Margaret Cooper K6WEK, Directors Sorin Constantinescu KN6YUH, Doug Teter KG6LWE, Walt Gyger K6WGY, and Darryl Presley KI6LDM. Also attending was Connie Stillinger W6EFI. A quorum was present. The meeting was called to order at 7:03pm

Officer Reports

- President's Report (Bob):
  - Congratulations to Doug Teter, our newest Life Member recipient!
  - I'm still working with Round Table Pizza to arrange for prompt service after the PAARA meeting.
  - I am reviewing different options on Zoom to give us maximum flexibility for starting the meeting.
- Secretary's Report (Ric):
  - We currently have 225 club members. 64 members have yet to renew for 2026. If you are one of these, renew today: See me at a club meeting, use the PayPal or Zelle links on the PAARA website, or send a check ($25.00) to PAARA, PO Box 911, Menlo Park CA 94026
  - We have renewed the FCC license for the club callsign W6ARA.
  - The California "Statement of Information" has been filed (required every two years).
- Treasurer's Report (Margaret):
  - Have filed the annual raffle report with the State.
  - Will soon demo GNUcash accounting system to the board.
  - Need to reconcile property values and close out 2025 books.
  - Will present financial results to the board.
  - Annual reports due to IRS and State of CA (due May 15).
- VP / Program Chair Report (Rob):

(Minutes — Continued on page 5)

(Minutes — Continued from page 4)

- Our speakers for February will be Rachel KK6DAC, and Mason KC3WNY. They will describe their work to set up an AREDN node at the Stanford Dish site.

Standing Items

- Attention: PAARA is your club. And PAARAgraphs is your newsletter. The articles in PAARAgraphs are your own. Please, take some time to tell your story about Ham Radio. Write it up, even briefly, and send to your editor KN6YGN@paara.org
  - This month we have an article on 30m antenna setup and surprises at Pacificon from Mikko Sannala AB6RF.
- Education Committee [Darryl]
  - We discussed the February question of the month.
  - Spring PAARA in the Park: Cupertino Memorial Park on April 18, 2026.
    - Our activity will be a multi-transmitter Fox Hunt. We will provide some handhelds and antennas for non-hams. We'll have small prizes for finding a fox: sticker, chocolate, etc.
    - We will invite club members to show their projects, kits, etc.
- IT Subcommittee Updates [Sorin]
  - Enhanced the Discord bot (https://github.com/PAARA-org/PAARAbot) to periodically read the member list and use it to track and report POTA/SOTA spots for them.
  - The last 10 members to join the club have been added to the announcements@paara.org distribution list.
- Bookkeeping Subcommittee Updates [Margaret]
  - 501c(3) status: After some discussion, we decided to contact other clubs for guidance. WVARA, for example, is a 501c(3) organization.
- PAARAGraphs Advertising [TBD]
  - It is time to send out invoices for 2026.
  - We need a volunteer to take over advertising management.

Old Business

- Life Membership Subcommittee update: The LM subcommittee recommended to the board that Doug Teter KG6LWE be awarded a life membership. A quorum of the board (not including Doug) approved this, and the award was given at the January club meeting. Congratulations, Doug!
- At the January meeting, we presented an overview of the club activities, membership, and financials. The board voted to make that presentation available to members on request. We will announce this in PAARAgraphs.
- N6NFI Repeater maintenance: Rolf N6NFI reports that he has this under control.
- Tower trailer maintenance: Doug will have a shop in San Martin quote on bearings lubrication and brakes checkup.

New Business

- 2026 is the ARRL "Year of the Club". We should submit photos of club activities, and perhaps the website, for consideration.
- T-shirt vendor: We are discussing with the T-shirt vendor ways to reduce the price of our Field Day T-shirts.

The meeting was adjourned at: 9:11 pm
Respectfully submitted

Ric Hulett N6AJS
PAARA Secretary

Discord Server

Do you want to talk to other PAARA members and friends about Homebrew, POTA, SOTA, digital modes, or any other ham radio topic? Download the App or use the Web version, then join our server by scanning the QR code, or visiting https://discord.gg/qPyn5h8p6H.

For any issues, please reach out to Sorin (KN6YUH) or Connie (W6EFI).

January Raffle Winners (left to right):
Ben (KN6UBF), Perry (KI6OPZ), Vicky (N7HBJ), Andy (KR6DD), Doug (KG6LWE)

PAARA RAFFLE

5th *Bioenno 12v, 12 Ah LiFePo4 battery*

4th *Vortex Optics Triumph HD 10x24 Binoculars*

3rd *POTA 15.5” x 31” desk mat*

2nd *DTAPE laser distance meter*

1st *Chicago Electric 5-watt solar panel*

HF Rig Loaner Program

The Palo Alto Amateur Radio Association (PAARA) offers an HF rig loaner program to help new General Class amateur radio operators get active on the HF bands. The program has two club-owned radios available for loan: a Yaesu FT-920 and an Icom IC-7300. The loan period is for six months, with the possibility of an extension if no one else is waiting to borrow a radio.

The loaner package includes the radio, a power supply, and a microphone. However, the borrower is responsible for providing their own antenna, tuner, and feedline. Any PAARA club member who is interested in borrowing a radio can get in touch with the program coordinators, Doug (KG6LWE) or Ric (N6AJS), by emailing radioloan@paara.org. For pictures and more, please check https://paara.org/loan.html.

Palo Alto Amateur Radio Association, Inc.

PO Box 911 Menlo Park, CA 94026

Officers

President .... Bob Ridenour, KN6YGN 650-575-4528 kn6ygn@paara.org
Vice President............. Rob Fenn, KC6TYD 650-888-9060 kc6tyd@paara.org
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Director ('26) .... Doug Teter, KG6LWE 650-743-7892 kg6lwe@paara.org
Director ('26) .... Darryl Presley, KI6LDM 650 255-2454 ki6ldm@paara.org

Appointed Positions

Membership .... Ric Hulett, N6AJS 408-332-4593 n6ajs@paara.org
Database..................... Ric Hulett, N6AJS 408-332-4593 n6ajs@paara.org
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Property Manager .... Doug Teter, KG6LWE
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Raffle Coordinator....... Rob Fenn, KC6TYD, kc6tyd@paara.org
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Webmasters................ webmaster@paara.org
Technical Coordinator . Christopher, AI6KG 408-348-0304 ch@murgatroid.com
QSL Manager.............. Ric Hulett, N6AJS 408-332-4593
Speaker Coordinator ... Rob Fenn, kc6tyd 650-888-9060

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Technical Tips .... Ric Hulett, N6AJS

VE Exams

De Anza Park, Sunnyvale, 2nd Saturday 10:30 am each month except November and December. See website for details and exceptions: http://amateur-radio.org

Electronics Flea Market (EFM)

Sponsorship: **A**ssociation of **S**ilicon **V**alley **A**mateur **R**adio **O**rganizations (ASVARO). The Electronics Flea Market is held at West Valley College, 14000 Fruitvale Ave, Saratoga.
Website: http://www.electronicsfleamarket.com/

PAARA — Palo Alto Amateur Radio Association

Meets 1st Friday 7:00pm each month at Room H-6, Cubberley Community Center; Net 145.230 - PL 100Hz Mondays at 8:30. See website at http://www.paara.org. For more information. contact: Bob Ridenour, KN6YGN, kn6ygn@paara.org, 650-575-4828

FARS — Foothills Amateur Radio Society

Meets 4th Friday each month at 7:00pm at Covington School, Los Altos.
Website: http://www.fars.k6ya.org

NCDXC — Northern California DX Club

Meets 3rd Thursday 7:00pm each month,
Repeater for member info 147.360. Contact president@ncdxc.org, Website: http://ncdxc.org. YouTube content: "The Northern California DX Club Official Channel". Cohost of the International DX Convention.

The 50MHz & Up Group of Northern California

This organization specializes in vhf + wak signal and microwave activities. Meetings are held on the first Tuesday of each month. Time is usually 5pm for in person meetings, and 7pm for Zoom only meetings. In person meetings are held Sports Basement, 1177 Kern Ave, Sunnyvale.
Always check the website, http://50MhzandUp.org, for correct information. Zoom information is also there.

San Mateo Radio Club W6UQ.ORG

Meets, 3rd Friday, January through November.
Tuesdays & Thursdays, [Directed] Net, 7pm, N6ZX 145.370Hz, -600KHz, PL107.2Hz
Contact: SanMateoRadioClub@gmail.com, Website: http://W6UQ.org/calendar

SPECS

Southern Peninsula Emergency Communication System users Group

Meets each Monday 7:30pm and 8:00pm.
See: https://specsnet.org/monday-night-net for more info.
Contact: https://www.specsnet.org/contact or board@specsnet.org

SCARES

South County Amateur Radio Emergency Service

Meets 3rd Thursday 7:30pm each month, Belmont EOC, Belmont City Hall, One Twin Pines Lane, Belmont CA 94002. Net is on 146.445 [PL 114.8] & 444.50 (PL-100) 7:30 Monday evenings. Contact: President mark Reynolds, KN6SRN
Web: http://k6mpn.org E-mail: pres@k6mpn.org

SCCARA

Santa Clara County Amateur Radio Association

Operates W6UU & W6UU/R, repeater 146.985-pl
Nets: 2m, 7:30pm Mon; 70cm, 10M (28.385) 8PM Thur.
Meets 2nd Mon each month @ 7:30 PM.
ARRL/VEC license testing contact 408-507-4698

SVECS — Silicon Valley Emergency Communications

Operates AA6BT repeater (146.115 MHz+)
Website: http://www.svecs.net or contact: Lou Stierer WA6QYS 408 241 7999

WVARA — West Valley Amateur Radio Association

W6PIY six-meter repeater on 52.58mHz. Normally, six-meters is linked with 147 and 223, while 441 and 1286 repeaters are linked.

Laptop Computers For Sale

These computers were primarily used by PAARA during Field Day and at the W1AW Special Event Station at Pacificon. We have recently upgraded to new laptops making these available to our club members for $50 each.

Specifications

Manufacturer:	Lenovo T240 41880FL6
Screen Size:	14 inch
System Type:	x64-based PC
Processor:	Intel® Core™ i5-2520M CPU @2.50GHz, 2 Cores
RAM:	8 GB
OS:	Microsoft Windows 7 Professional

PAARA Badges

Badges can be ordered through our website https://paara.org/badges.html. Scroll to the bottom of the page and fill out the info. All badges will be mailed.

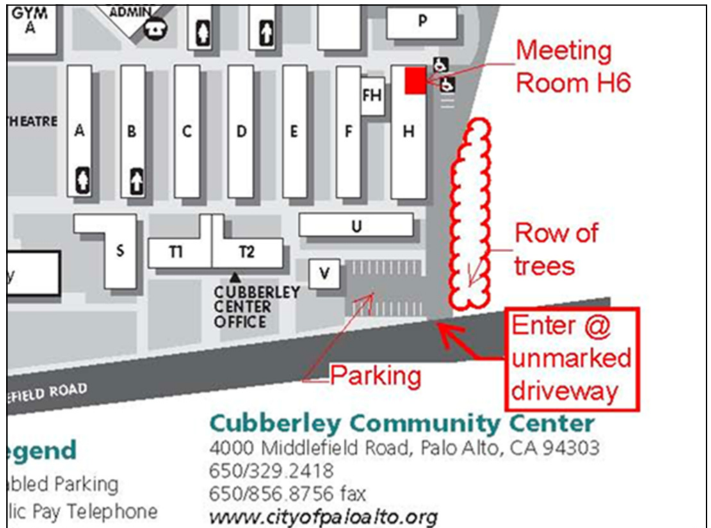
The cost for a badge is $30.00.

PAARA Weekly Radio Net

Info and Swap Session
every Monday evening at 8:30pm
on the N6NFI 145.230 MHz repeater

Week	Control Operator
1st	Doug - KG6LWE
2nd	Doug - KG6LWE
3rd	Ric - N6AJS
4th	Rob - KC6TYD
5th	Rob - KC6TYD

If you're interested in trying out at Net Control, Contact Doug, KG6LWE. It's good practice, and lots o' fun! Give it a try.



Meeting Location — Middlefield Road between San Antonio and Charleston in Palo Alto. 4000 Middlefield Road

Palo Alto Amateur Radio Association

P.O. Box 911, Menlo Park

California 94026-0911

Club meetings are on the first Friday of each month, 7:00pm at the Room H-6, Cubberley Community Center.

Radio NET & Swap Session every Monday evening, at 8:30pm, on the 145.230 –600 MHz repeater, PL 100Hz.

Membership in PAARA is $25.00 per calendar year, which includes one subscription to PAARAgraphs
$6 for each additional family member (no newsletter).

Make payment to the

Palo Alto Amateur Radio Association,
P.O. Box 911, Menlo Park, CA 94026-0911

ARV'S, WA6UUT (SK)

WEDNESDAY

HAM RADIO

LUNCHEON

Our 19th year!

- Since May 2, 2007 –

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Sunnyvale, California

415 East El Camino Real

(Just "North" of South Fair Oaks Avenue on El Camino Real)

11:30 AM ~ 3:00 PM

Website:www.blackbeardiner.com

Many food choices available from the breakfast, lunch or dinner menus. Ample parking is available. Walk in & "bear" left for our location in

Submit items to **PAARA*graphs*** by the 3rd Wed to: paaragraphs@paara.org

Text: .doc, .rtf, or .txt

Photos:

jpg, png or tiff

Subscription Problems?

Contact Database Manager:

Ric Hulett, N6AJS

PAARA W6OTX Repeaters

VHF DMR	144.9625 MHz +2.5 MHz CC3	Slot 1: Dynamic Slot 2: NorCal BM
UHF DMR	444.475 MHz +5 MHz CC1	Slot 1: Dynamic Slot 2: NorCal BM
33cm FM	927.075 MHz -25 MHz PL 107.2 Hz	AllstarLink Node 65314 Echolink: W6OTX-R

PAARA*graphs* Ad Rates

PAARA*graphs* accepts paid advertisements from non-members. (short personal ads remain free for members in good standing). **All ad rates listed are per issue**.

1. Not-for-profit ads by association members for ham-related items and wants. No cost for business card–size ads (additional space at $2.50 per business card size per issue).
2. For Profit organizations and/or individuals: $5-business card size, $14.00-quarter page, $25-half page, $50 full page or back cover per issue.

PAARA*graphs* — February 2026

STAMP

Palo Alto Amateur Radio Association, Inc.
PAARA*graphs* Newsletter
P.O. Box 911

FIRST CLASS MAIL

Address Service Requested